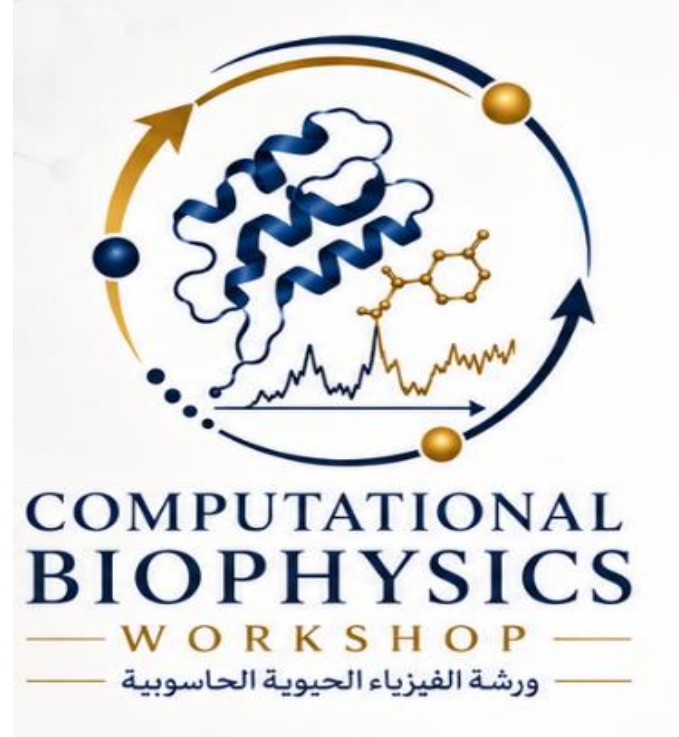


Molecular Dynamics Simulation

What happens after molecular docking?



What is after molecular docking?

Docking gives you a single snapshot — a static pose of one ligand in one receptor. But proteins and ligands are not static. They wiggle, breathe, bend, and rearrange.

THE QUESTION

Docking hands you a frozen picture. Reality is a movie. To understand how a drug actually binds, releases, or causes a protein to change shape, we need to watch the system move over time.

THE ANSWER

Molecular Dynamics Simulation (MDS) numerically solves Newton's equations of motion for every atom in the system — protein, ligand, water, ions — and produces a trajectory: a movie of the molecule in motion over nanoseconds to microseconds.

WHAT YOU GET

- A trajectory file — the 3D coordinates of every atom at every time step
- Insights into flexibility, conformational changes, and binding stability
- Quantities like RMSD, RMSF, free energy, and hydrogen-bond lifetimes

“

*“If we were to name the most powerful assumption of all, which leads one on and on in an attempt to understand life, it is that **all things are made of atoms**, and that everything that living things do can be understood in terms of the **jiggings and wiggings of atoms**.”*

— Richard P. Feynman (1918–1988)

Today's roadmap

Four chapters — from the foundations of structure and force fields, to the simulation loop, and what MD can and cannot do.

01



Foundations

What is MDS, Obtaining structures, the PDB format, and what makes a molecule move.

CHAPTER 01

02



Force fields

The energy equation that controls all motion, and where its parameters come from.

CHAPTER 02

03



The simulation loop

Newton's equations, integrators, initial velocities, and time step selection.

CHAPTER 03

04



Applications & limits

What MD is good for, where it fails, and what you need to run one.

CHAPTER 04

CHAPTER 01

Foundations

What is **MDS**?

Three words — each one matters.

01



Molecular •

The system we simulate is a molecule — usually a protein, a ligand, a membrane, or any combination of these. The molecule is described as a collection of atoms, each with a position, mass, and charge. Together, these atoms define the system to be simulated.

02



Dynamics •

The molecule is not static — its atoms move, vibrate, rotate, and translate in 3D space. Dynamics is the study of how these motions evolve over time under the influence of forces. The forces come from chemical bonds, electrostatics, and van der Waals interactions.

03

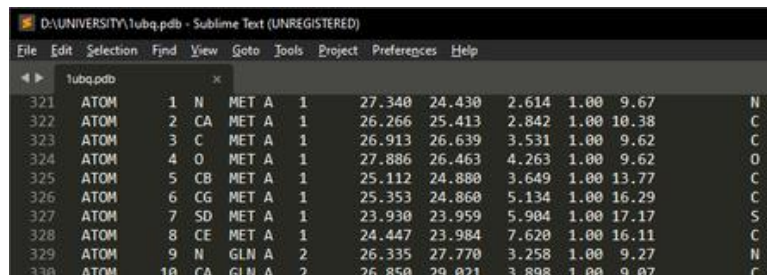


Simulation •

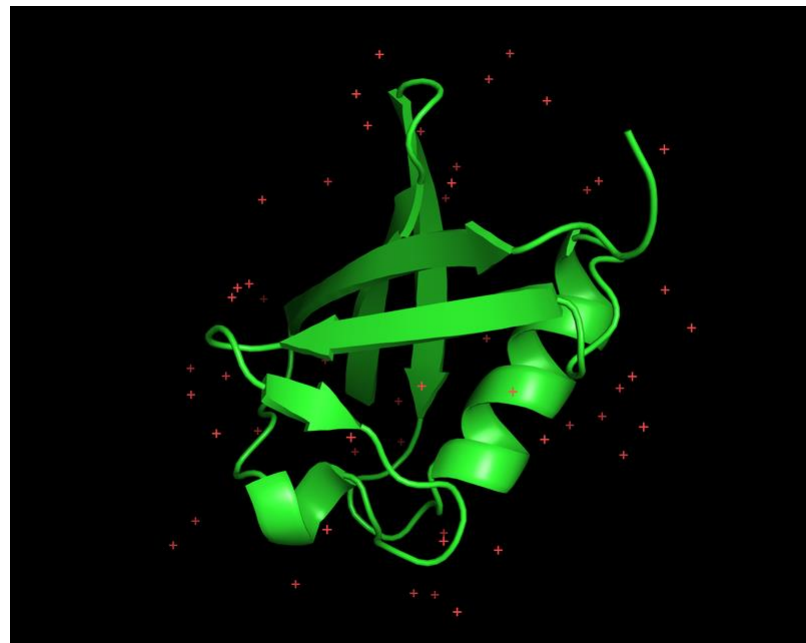
A simulation is a computer experiment — we numerically solve the equations of motion for every atom at small time steps and produce a trajectory. Each step gives us the new positions and velocities of all atoms, saved as a movie of the molecule in motion.

How do we obtain a 3D structure and view it ?

Two routes — experimental and theoretical — both ending in a .pdb file we can view using different programs.



321	ATOM	1	N	MET	A	1	27.340	24.430	2.614	1.00	9.67	N
322	ATOM	2	CA	MET	A	1	26.266	25.413	2.842	1.00	10.38	C
323	ATOM	3	C	MET	A	1	26.913	26.639	3.531	1.00	9.62	C
324	ATOM	4	O	MET	A	1	27.886	26.463	4.263	1.00	9.62	O
325	ATOM	5	CB	MET	A	1	25.112	24.880	3.649	1.00	13.77	C
326	ATOM	6	CG	MET	A	1	25.353	24.860	5.134	1.00	16.29	C
327	ATOM	7	SD	MET	A	1	23.930	23.959	5.904	1.00	17.17	S
328	ATOM	8	CE	MET	A	1	24.447	23.984	7.620	1.00	16.11	C
329	ATOM	9	N	GLN	A	2	26.335	27.770	3.258	1.00	9.27	N
330	ATOM	10	CA	GLN	A	2	26.850	29.021	3.898	1.00	9.07	C



I have a structure — how can I make it **move**?

A static structure gives you positions.

Add velocities, advance time in small steps, and save coordinates — you get a movie.

⊕ THE RECIPE

You start from one static conformation — a single set of atom positions (e.g. from docking, X-ray, or AlphaFold). To make it move, you assign a velocity to every atom, then advance the simulation in tiny time steps.

► WHAT YOU GET

- After each time step, save the new 3D coordinates of every atom
- Stack these snapshots together over millions of steps
- The result is a trajectory — a movie of the molecule in motion

☰ KEY TERMS

Trajectory = a sequence of coordinate snapshots over time

Frame = one snapshot in the trajectory

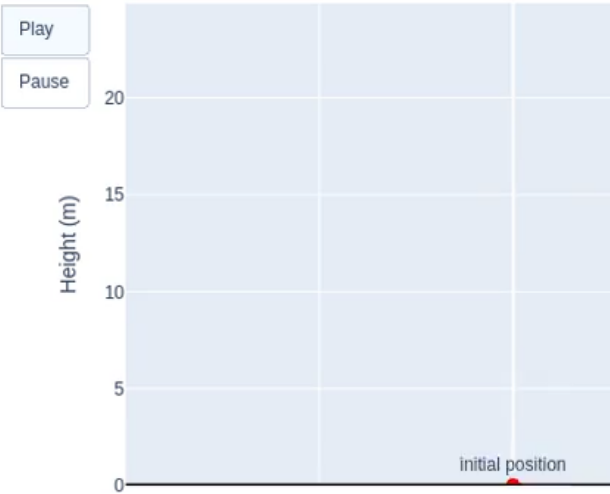
Time step = how far time advances between snapshots (typically 1–2 fs)

CHAPTER 02

The Force Field

The mathematical function that decides how atoms push and pull each other — the engine of every MD simulation.

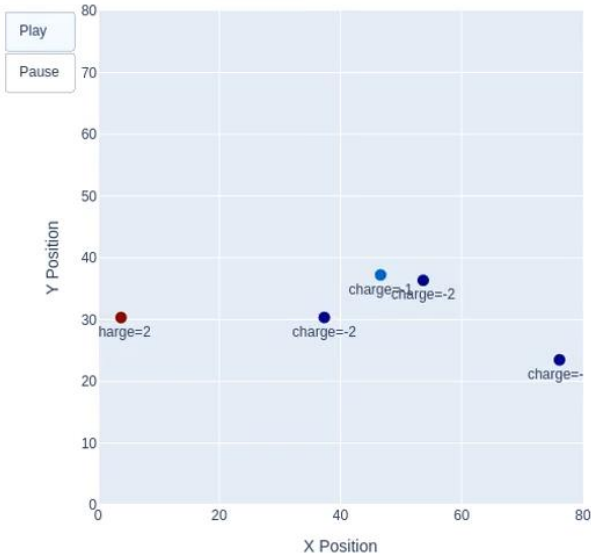
Bouncing Ball Simulation with Elastic Collisions



Bouncing Ball Simulation with inElastic Collisions



Particle Simulation with Coulomb Force (Different Charges)



The force field — the **energy equation**

A single mathematical function that gives the total potential energy of the system for any 3D arrangement of atoms.

TOTAL POTENTIAL ENERGY

$$\begin{aligned}
 U(\mathbf{r}^N) = & \sum_{\text{bonds } i} a_i (d_i - d_{i,0})^2 \\
 & + \sum_{\text{angles } j} b_j (\theta_j - \theta_{j,0})^2 \\
 & + \sum_{\text{torsions } k} \left[\sum_n c_{k,n} [1 + \cos(\omega_k n + \gamma_k)] \right] \\
 & + \sum_{\text{pairs } ij} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + 4\epsilon_{ij} \left[\left(\frac{r_{ij}}{\sigma_{ij}} \right)^{-12} - \left(\frac{r_{ij}}{\sigma_{ij}} \right)^{-6} \right]
 \end{aligned}$$

♦ **Bond stretching**

Spring-like energy from stretching or compressing a bond.

♦ **Angle bending**

Energy from bending the angle between three bonded atoms away from its equilibrium θ_0 .

∪ **Torsional rotation**

Energy from rotating around a bond — periodic in the dihedral angle φ . Captures the barrier between conformations.

♦ **Non-bonded**

Two terms: electrostatic (Coulomb) between charged atoms, and van der Waals (Lennard-Jones) for uncharged atoms.

Where do all these **parameters** come from?

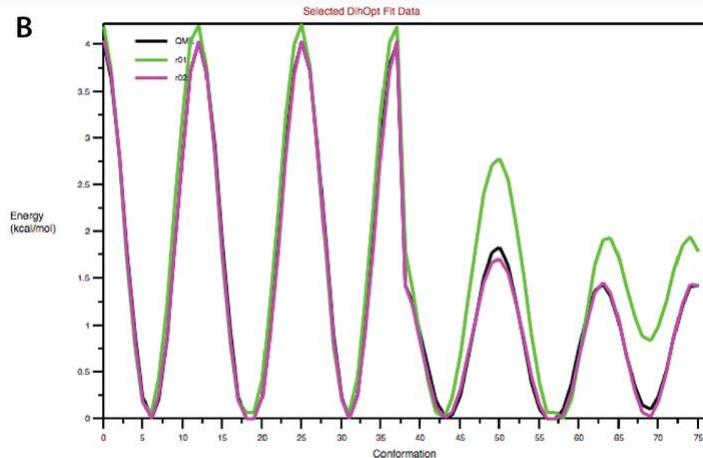
Every k_b , r_0 , k_θ , θ_0 , etc... in the force field must have a numerical value. Where do they come from?

🔗 TWO SOURCES

Force field parameters come from two complementary sources: **quantum mechanics (QM)** calculations on small molecules, and **experimental measurements**. Because both are used, force fields are called **semi-empirical**.

⚙️ HOW IT WORKS — TORSION EXAMPLE

For each torsion angle, run a QM calculation at many angle values to get the energy curve. Then adjust the force field's parameters until the FF curve matches the QM curve as closely as possible. The same fitting is done for bond, angle, and partial-charge parameters.



☰ POPULAR FORCE FIELDS

CHARMM36

AMBER (ff14SB, ff19SB)

GROMOS

OPLS

CHAPTER 03

The Simulation Loop

What happens **behind the scenes**?

Every MD step is just Newton's second law — applied to every atom, over and over.

◆ THE FOUNDATION

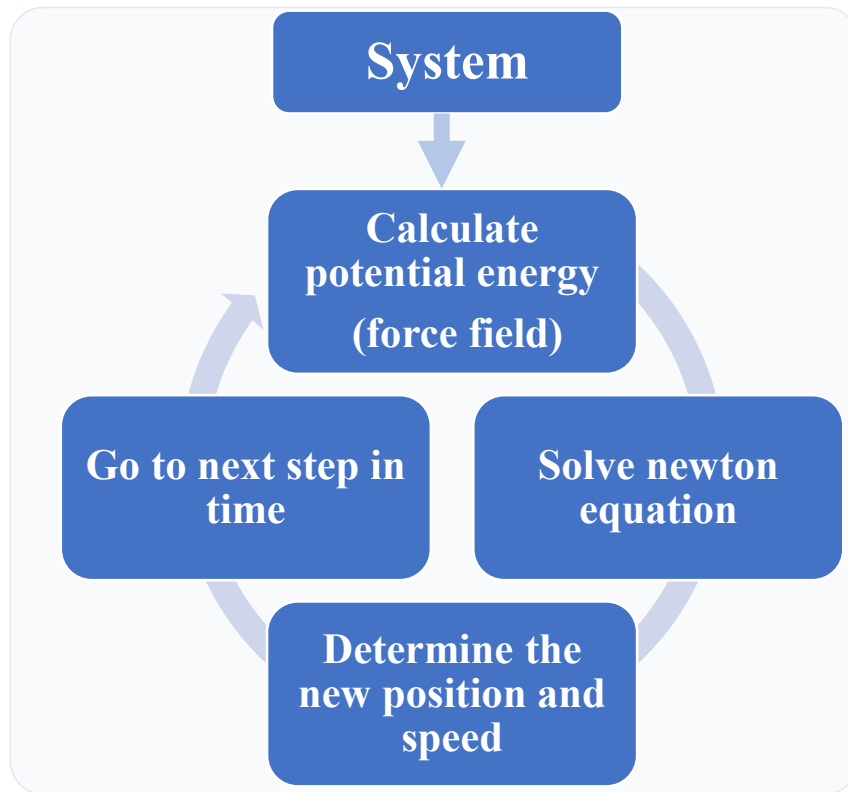
Newton's second law of motion says force equals mass times acceleration. In MD, this becomes $\mathbf{F}_i = \mathbf{m}_i \cdot \mathbf{a}_i$ for every atom i in the system.

⚙️ WHAT EACH PIECE MEANS

- $\mathbf{F}_i$ — the total force on atom i , computed from the force field
- $\mathbf{m}_i$ — the mass of atom i (e.g. carbon ≈ 12 Da, hydrogen ≈ 1 Da)
- $\mathbf{a}_i$ — the resulting acceleration of atom i

⚡ THE FORCE

The force on each atom comes from the negative gradient of the potential energy V with respect to that atom's position: $\mathbf{F}_i = -\partial V / \partial \mathbf{r}_i$. The force field gives you V ; differentiating it gives you the force on every atom in one shot.



How do we **integrate** Newton's equations?

Newton gives us accelerations. To get positions over time, we need an integrator — and one of the most famous is Velocity Verlet.

♦ WHAT IS AN INTEGRATOR?

An integrator is a numerical recipe that takes the current positions, velocities, and accelerations of every atom, and computes the new positions and velocities one small time step Δt later. Repeat millions of times and you get a trajectory.

$$r(t + \delta t) = r(t) + v(t)\delta t + \frac{f(t)}{2m} \delta t^2$$
$$v(t + \delta t) = v(t) + \frac{f(t + \delta t) + f(t)}{2m} \delta t$$

But how do I get the **initial velocities**?

You start from one static structure — every atom has a position but no velocity. To kick off the simulation, you need to assign each atom an initial velocity.

? THE PROBLEM

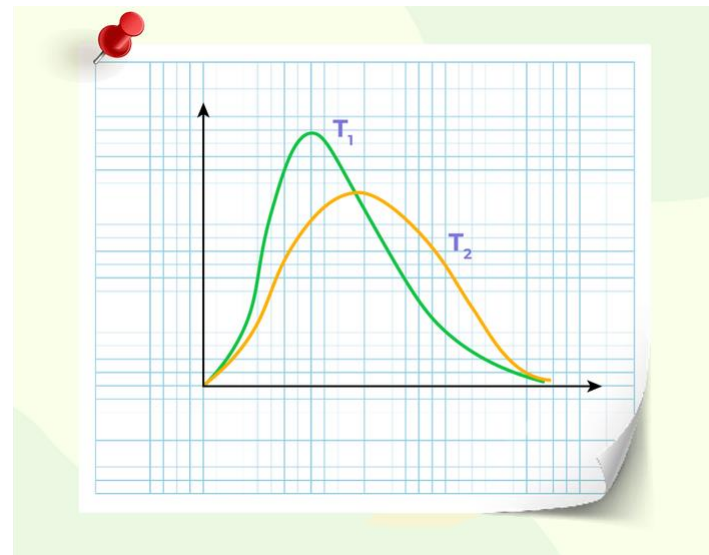
Your input structure (e.g. a docked complex) contains atom positions but no velocities. Without velocities, the integrator has nothing to advance. We need a physical way to assign them.

✂ THE SOLUTION — MAXWELL-BOLTZMANN

The Maxwell-Boltzmann distribution relates the average kinetic energy of atoms to the temperature of the system. At a chosen temperature T , the distribution tells us how likely each velocity value is. We sample each atom's velocity from this distribution.

✓ IN PRACTICE

- Pick a target temperature T (e.g. 300 K)
- For each atom, sample v_x , v_y , v_z from the Maxwell-Boltzmann distribution



Maxwell-Boltzmann velocity distribution — at higher T , atoms move faster on average.

After assignment, the system is coupled to a thermostat (e.g. V-rescale, Berendsen) to keep T constant during equilibration.

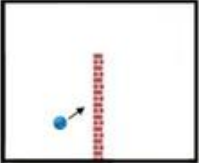
Choosing the time step Δt

Too large $\rightarrow$ the simulation explodes. Too small $\rightarrow$ you waste computation. The right choice depends on the fastest motion in your system.

LAMMPS Workshop 2025 - Day 1 - Tutorial

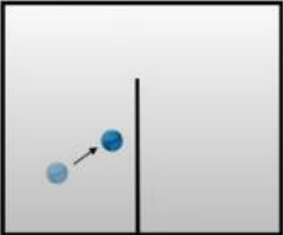
A Limitation of Computers: Everything is Discrete

Let's put a very thin brick wall into our box



In an MD simulation, particles must move a finite distance on every time step.

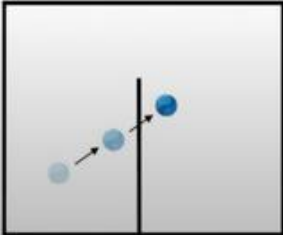
Due to discrete time step, particles 'teleport' in space between time steps.



Therefore, our marble could pass through a sufficiently thin (even if very solid) brick wall.

$\longrightarrow$

This issue is the cause of most instabilities during MD simulations.



 a y
 Created by Eam

▶ ⏮ 🔊 39:51 / 7:57:55
⏭ ⏮ 🔊

Choosing the time step Δt

Too large $\rightarrow$ the simulation explodes. Too small $\rightarrow$ you waste computation. The right choice depends on the fastest motion in your system.

☰ THE RULE OF THUMB

The time step must be small enough to resolve the fastest motion in the system. The fastest vibrations in biomolecules are X–H bond stretches (N–H, O–H, C–H) — period ~ 10 fs. A safe time step is roughly **1/10 of the fastest period**.

⚙️ TWO PRACTICAL CHOICES

$\Delta t = 1$ fs Flexible bonds

All bonds are flexible. The fastest X–H vibrations are resolved. Safest, but you need **1,000,000 steps** for 1 ns of simulation time.

$\Delta t = 2$ fs Bonds constrained

Use algorithms like **LINCS** or **SHAKE** to freeze X–H bonds. The fastest remaining motions are heavier-atom vibrations; 2 fs is now safe. Halves the number of steps.

♦ STEPS $\rightarrow$ TIME

1 fs $\times$ 1,000,000 steps = 1 ns

2 fs $\times$ 500,000 steps = 1 ns

2 fs $\times$ 5,000,000 steps = 10 ns

2 fs $\times$??? steps = 100 ns

CHAPTER 04

Applications & Limitations

Is MD simulation **important**?

Yes — MD tells you how a biomolecular system responds to a perturbation. Five classic use cases.

A

Remove a ligand

Take an experimentally determined protein-ligand complex, remove the ligand, and watch the dynamics of the apo protein (the protein without ligand). Reveals which parts of the pocket collapse or rearrange when the drug leaves.

B

Replace a ligand

Swap the bound ligand for a different one — exactly what docking predicts. MD then shows whether the new ligand stays put, induces different conformations, or destabilizes the protein. Connects docking to dynamics.

C

Mutate residues

Change one or more amino acids in the protein (e.g. simulate a known drug-resistance mutation) and observe how the dynamics and binding pocket shift. Used to predict resistance before it appears in the clinic.

D

Apply external forces

Pull on a specific part of the protein (steered MD) to identify key residues responsible for a function — folding, unfolding, or mechanical signaling. Maps mechanical coupling inside the structure.

E

Change environment

Change the molecular environment of the simulated protein — temperature, pH (via protonation), salt concentration, or membrane composition. Reveals how environment modulates function and stability.

Is classical MD simulation **perfect**?

No. Classical MD has three well-known limitations — and three known workarounds.

01 Force field accuracy

LIMITATION

MD is only as good as its force field. Better FFs → better results. Different FFs (CHARMM vs AMBER vs OPLS) can give different answers for the same system.

WORKAROUND

Use the latest FF version (CHARMM36m, AMBER ff19SB) and validate with experimental observables.

02 Limited sampling

LIMITATION

Is your simulation long enough to see all relevant conformations? Generally, the answer is No — many biologically relevant motions take microseconds to milliseconds, far beyond routine MD.

WORKAROUND

Use enhanced sampling methods: Replica Exchange MD (REMD), Metadynamics, Gaussian Accelerated MD (GaMD).

03 Cannot break bonds

LIMITATION

Classical MD uses fixed bonds — it cannot simulate enzymatic reactions where bonds form or break. You see conformational changes only, not chemistry.

WORKAROUND

Use QM/MM (Quantum Mechanics / Molecular Mechanics) — treat the active site quantum-mechanically and the rest classically.

What do you need to run an MD simulation?

Four ingredients

01 Hardware



A modern multi-core CPU is the minimum. For speed, a GPU (NVIDIA with CUDA) is strongly recommended — GPU MD is **10–100× faster** than CPU-only.

⚡ Example: 1 × RTX 4090 $\approx$ 200 ns/day for a 25k-atom system

03 Simulation software



Pick a MD engine. GROMACS is free and the fastest on GPU; NAMD and OpenMM are also popular; AMBER and CHARMM have their own packages; Desmond is commercial.

◆ GROMACS · NAMD · AMBER · OpenMM · Desmond

02 Force field + input files



A force field choice (AMBER, CHARMM, OPLS) plus a topology file (lists every atom, bond, angle) and a configuration / MDP file (run parameters: Temperature, Pressure, time step, nsteps).

📄 Files: topol.top · conf.mdp · coords.gro

04 Solvated + neutralized system



The protein or complex must be placed in a box of water (explicit solvent), and ions (Na^+ , Cl^-) added to neutralize the system and reach physiological salt concentration (~ 0.15 M).

◆ Use CHARMM-GUI to prepare everything in one go

Programs & resources

Everything you need to run your first MD simulation — all free.

CHARMM-GUI

Web interface to prepare your entire MD system — solvation, neutralization, FF parameters, GROMACS/NAMD input files.

 <https://charmm-gui.org/>

GROMACS

The MD engine itself — free, open-source, GPU-accelerated. The fastest MD code on commodity hardware.

 <https://www.gromacs.org/>

Colab — Run MD in the cloud

Pre-built Google Colab notebook to run an MD simulation end-to-end in the browser — no local install.

 <https://colab.research.google.com/drive/1GrBoF17OJVnT633oW9VfY5KiMwIC8PYA?usp=sharing>

Colab — Trajectory analysis

Companion Colab notebook for analyzing MD trajectories.

 <https://colab.research.google.com/drive/1i1-57oPl5FGuR1MmoSewHjVlwls2f-F?usp=sharing>

✦ You now have everything to run your first molecular dynamics simulation. EXCEPT ONE THING!

THE GIFT

(صندوق النمر)

